
IMAGE: STEPHAN LUDEWIG
FULL THRUST
LIGHT
GZG
GROUND ZERO GAMES
MECHWORLD
CLASSIC BATTLETECH CHAPTER LEAGUE



IMAGE: SASCHA KEUNECKE

WHAT IS FULL THRUST LIGHT?

Full Thrust has been one of the most consistently popular Starship Combat wargames around, ever since the first edition was published way back in 1991.

The rules set presented here, which we call **FULL THRUST LIGHT**, is a stripped-to-the-basics version that introduces the most important core rules of the game, and is ideal for getting new players into the system. **FULL THRUST LIGHT** uses only the simplest movement rules (for "cinematic" style play) and just a couple of the most common weapon systems. The "complete" version of **FULL THRUST** (currently in Second Edition with add-ons, which we refer to as "FT2.5", with a Third Edition under development) introduces much, much more – loads more weapon options, defensive systems and screens, fighter operations, an optional "vector" movement system, and a complete design-your-own-ships construction and costing system.

So, enjoy this simple free intro to the rules, then check out our website at **www.gzg.com** for all the rest!

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WHAT YOU'LL NEED:

It is possible to play **FULL THRUST** with card counters representing the starships, but for the full miniatures gaming experience you'll ideally want models. Our extensive **FULL THRUST** miniatures line can provide you with everything you need, but you can also use models from any other manufacturers. If you are using one of our **FT INTRO PACK** sets, you'll have two good starter forces of 8 ships per side ranging in size from small Frigates up to a Battlecruiser each, which is an ideal set-up to get you into the game.

All ship models should be mounted on stands, and all distances and ranges are measured to and from the **CENTRE** of the model's stand.

Each player in the game will need a Ship Record Sheet filled out for his fleet, for plotting movement orders and recording damage etc. You'll also need a handful of ordinary six-sided DICE, and a tape measure or long ruler marked in your chosen Measurement Units (see below).

Finally, you'll need a flat playing surface; **FULL THRUST** does not require a hexagon or square grid map, just a plain tabletop - though your playing area can be made much

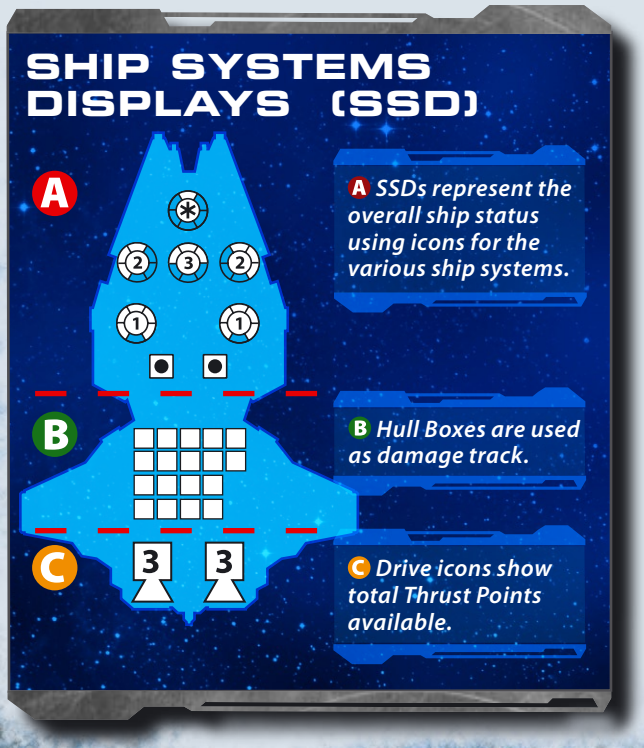
more attractive by using a black cloth or board speckled with a "star-field" of paint dots.

What you need to play: Models or Markers representing ships, record sheets, some six sided dice, ruler or tape measure and a playing area.

SHIP RECORD SHEETS:

The Ship Record Sheets that you'll find on the reverse of these rules are already set up for some simple generic ship designs based on the selection of miniatures in the intro Pack, so you can start to play almost immediately - just photocopy one sheet per player and you are ready to go. The two sheets each have different ship designs to give you some variety, but you can use identical sheets for both fleets if you prefer. It doesn't matter what miniatures you use with these designs, and after your first game or two you should feel free to modify the designs by swapping some weapons and systems around to suit your own preferences. Permission is hereby granted for the Ship Record Sheets to be photocopied or printed off as desired for non-commercial purposes.

The Ship Record Sheet is divided into two parts: the Ship Systems Displays (SSDs), one for each ship in your fleet, and the Movement Plot which is used for writing the movement orders for each game turn.



Each ship uses one line on the Movement Plot for recording its movement orders; the first (greyed-out) line on the plot is filled in with some example orders to illustrate how everything should be recorded.

The individual SSDs for each ship have a number of symbols, or **ICONS**, that represent the various **SYSTEMS** (drives, weapons etc) that the ship carries. The far right SSD on the sheet - the greyed-out one - should not be used in play; it is provided as a reference example to explain all the icons.

⚠ **IMPORTANT:** ***Please note that the ship designs given in the SSDs here are NOT "full" FT designs – to avoid confusion, they only show the systems that are relevant to the introductory rules presented here.***

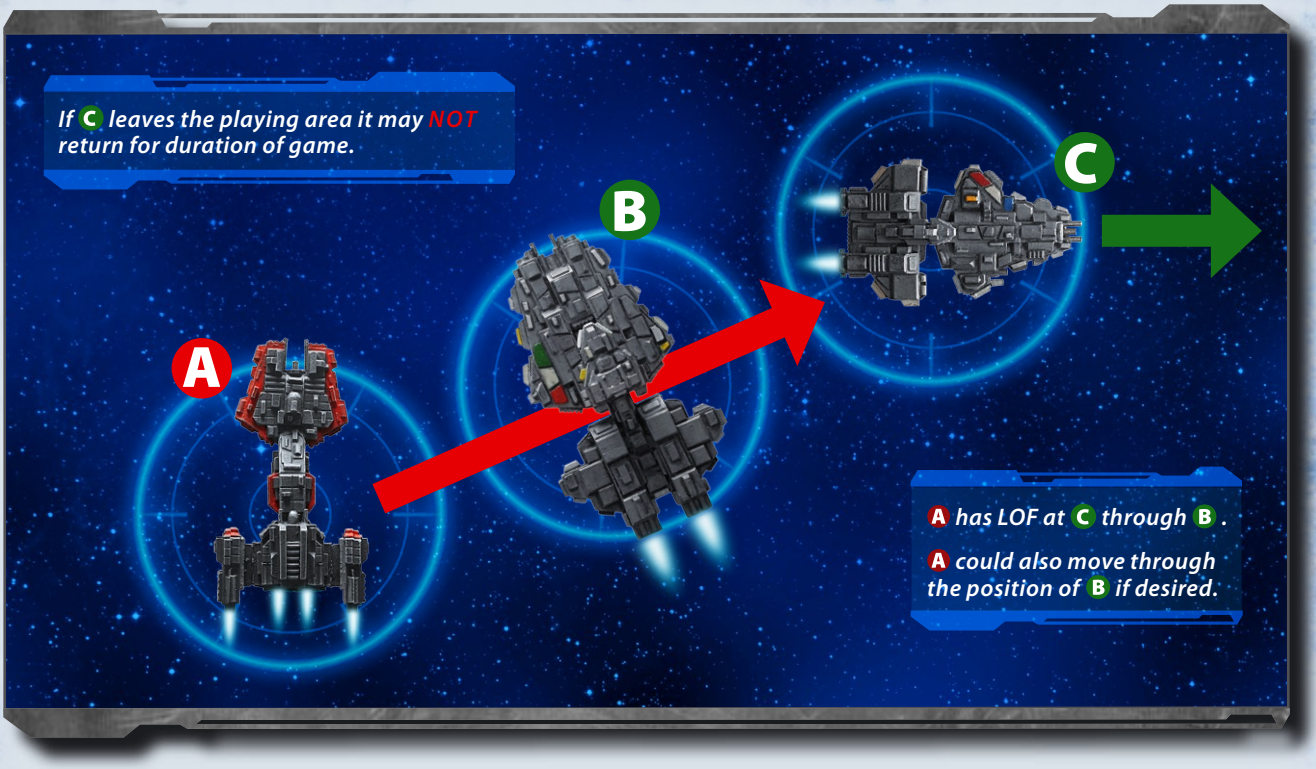
SSDs represent the overall ship status using icons for the various ship systems. Hull Boxes are used as damage track. Use movement plot to record your movement orders.

DISTANCE AND MEASUREMENT:

All distances in FULL THRUST are given in "Measurement Units" or mu; you can choose what one mu represents, to suit the size of your playing area. The standard that we normally use is 1mu = 1 inch, which works best if you have a fairly large table, but if you want more action in a smaller space then feel free to use 1mu = 1 centimetre.

Ships never block line-of-fire from each other, and may freely move through/across other ships during the movement phase; space is, after all, three dimensional even though we choose to represent it two-dimensionally here for simplicity of play. If there is a conflict of placement between actual miniatures at the end of the movement phase, due to the physical size of the models, then they should be placed as closely as possible to their intended positions. This should be with the agreement of all players, displacing smaller ships in preference to larger ones and maintaining courses as the highest priority. Full Thrust Light has no provision for collisions or deliberate ramming, but this is covered in the full FT rules!

If a ship leaves the playing area, either deliberately or through player error in plotting movement orders, in basic terms it is out of the game and may NOT return. This rule should be considered flexible, however, and returning to play MAY be allowed with the agreement of all involved provided that this is decided at the start of the game.



Rules refer to distances in mu; 1 mu = your favourite unit (inch, cm). Distances and ranges are measured to and from the CENTRE of the model's stand. Ships never block LOF and movement and there are no collisions or ramming; Ships leaving the table may not return to battle in the basic game.

COURSES AND FIRE ARCS:

The direction in which a ship is moving is referred to as its **COURSE**. In the simple "cinematic" movement system here, ship models must always face in the direction they are moving. There are 12 possible Courses, denoted by a "clockface" with a fixed orientation on the playing area (ie: Course 12 should be defined as pointing directly towards one table edge, and all courses are then defined relative to that). Thus a ship moving directly in the "12" direction would be on Course 12, while one traveling in the opposite direction would be on Course 6. The "clockface" is also used to define the **FIRE ARCS** of a ship's weapons, though the fire arcs are relative to the SHIP rather than relative to the playing area. Ships in Full Thrust Light have 6 fire arcs of 60 degrees each; working clockwise, there is the **FORE (F)** arc from 11 o'clock to 1 o'clock, the **FORE STARBOARD (FS)** arc from 1 to 3 o'clock, the **AFT STARBOARD (AS)** arc from 3 to 5 o'clock, the **AFT (A)** arc from 5 to 7, **AFT PORT (AP)** from 7 to 9 and finally the **FORE PORT (FP)** from 9 to 11 o'clock.

Each weapon system icon on the SSD has a ring of 6 segments round it that correspond with the 6 fire arcs – those arcs through which the weapon may be fired are left





IMAGE: SASCHA KEUNECKE

clear, while arcs through which it may **NOT** fire are filled in black. A weapon may only be fired at a target that is within a valid fire arc of the firing ship during the firing phase of the turn. Even if a ship passed through a valid arc during the movement phase, if it is out of arc by the firing phase it may not be attacked.

⚠ **IMPORTANT:** ***It will be noted that all weapon icons on the sample SSDs have the AFT (A) arc filled in, even if they can fire through all 5 other arcs; in Full Thrust Light, NO weapon may be fired out of the AFT (A) arc of any ship. Thus the aft 60 degree arc of any ship is a weapons "blind spot", though of course incoming enemy fire may hit the ship through this arc exactly as through any other.***

⛔ *COURSE is the direction of movement; there are 12 possible courses indicated by a "clockface" with fixed orientation; Firing arcs are using a similar "clockface" but relative to the ship model, indicated by a ring of 6 arcs surrounding each weapon icon on the SSD.*

SEQUENCE OF PLAY:

⚠ ***The sequence for a game turn of Full Thrust Light is as follows:***

PHASE 1:
ORDER PLOTTING PHASE

PHASE 2:
SHIP MOVEMENT PHASE

PHASE 3:
SHIP WEAPONS FIRE PHASE

Once all players are happy that everything they need to do in the turn is completed, the next game turn begins with a new order plotting phase. The full version of the game introduces a number of extra phases to the turn, but for now the three basic ones here are all you'll need.

⛔ *A game turn breaks down into Order Plottting, Ship Movement and Weapons Fire Phases.*

ORDER PLOTTING PHASE:

In this phase of the turn, all players must write **MOVEMENT ORDERS** for each of the ships in their fleet. Any ship that does not have an order written for it will continue to move on the same course and at the same velocity as in the previous turn. A ship's movement order is usually written in an abbreviated notation as explained below, and must specify any change in velocity (acceleration or deceleration) and/or change in course (to Port or Starboard, and by what amount) that the ship is to perform in this game turn.

After any changes to velocity or course are plotted, the new total velocity is written into the next empty "**V**" box on the order plot, and the new course heading is written in the "**C**" box, to show the new starting velocity and course for the next turn.

All ships have one or more **DRIVE SYSTEM** icons on their SSD, which represent their "sublight drive" engines. Each drive icon has a number which shows the **THRUST POINTS** that it provides; a ship's total available thrust per game turn is the total of the numbers in all of its functioning drives. A ship that loses all its drive system(s) due to damage is no longer able to change course OR velocity, and will continue to travel at its current course and velocity until it exits the playing area.

⚠ **IMPORTANT:** ***In Full Thrust Light, all ships may only use up to HALF (rounded UP) of their total available thrust points for changing course in any game turn, and any thrust points thus used may NOT also be used for acceleration or deceleration at the same time. The total amount of thrust points applied in one game turn cannot exceed the total thrust points available from the ship's drives.***

For example, a ship with Thrust 6 drives could only apply up to 3 of those thrust points for changing course, but it could then use any remaining points for changing velocity in the same game turn if desired. Thus an order plot of "+4, P2" (4mu acceleration, with a 2-point turn to Port) would be allowed, but one of "+2, P4" would not – although 6 thrust points are available in total, the ship is not allowed to use more than 3 of them for course changes.

Orders for ships must be written down using the movement plot on the ships SSD. Ships without orders continue to move on last course and velocity; Thrust points on the SSD indicate the maximum thrust for acceleration/deceleration and course changes. Up to half of total available thrust points may be used to change course.

Course plot abbreviations:
Acceleration: e.g. +4
Deceleration: e.g. -3
Port turn: e.g. P2
Starboard turn: e.g. S3
Velocity at end of movement: V
Course at end of movement: C

MOVEMENT PHASE:

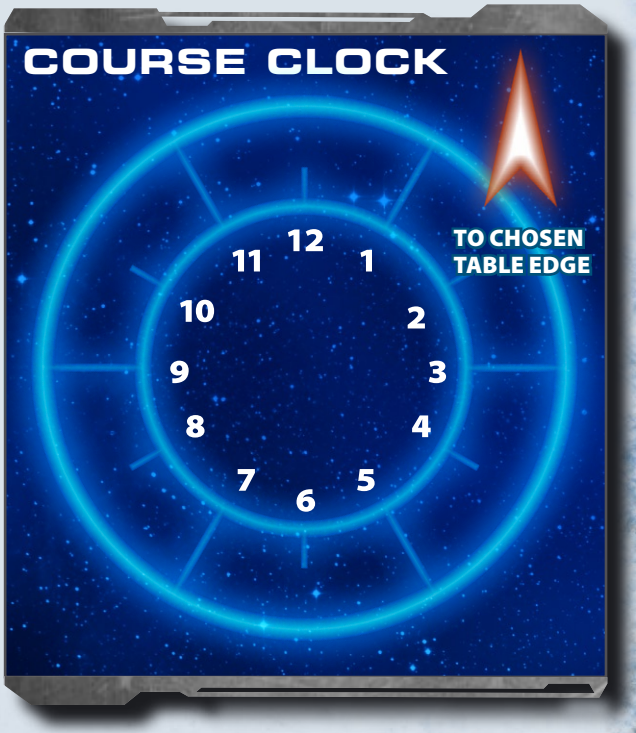
In the movement phase, each player moves his ship models in exact accordance with the orders he has just written for them in the plotting phase. Both/all players may move their models simultaneously if they wish, or one may move followed by the other – but as both will be following their written orders it will not make any difference; neither player may modify their ordered movement in any way to react to the other's moves.

Execute movement according to written orders.

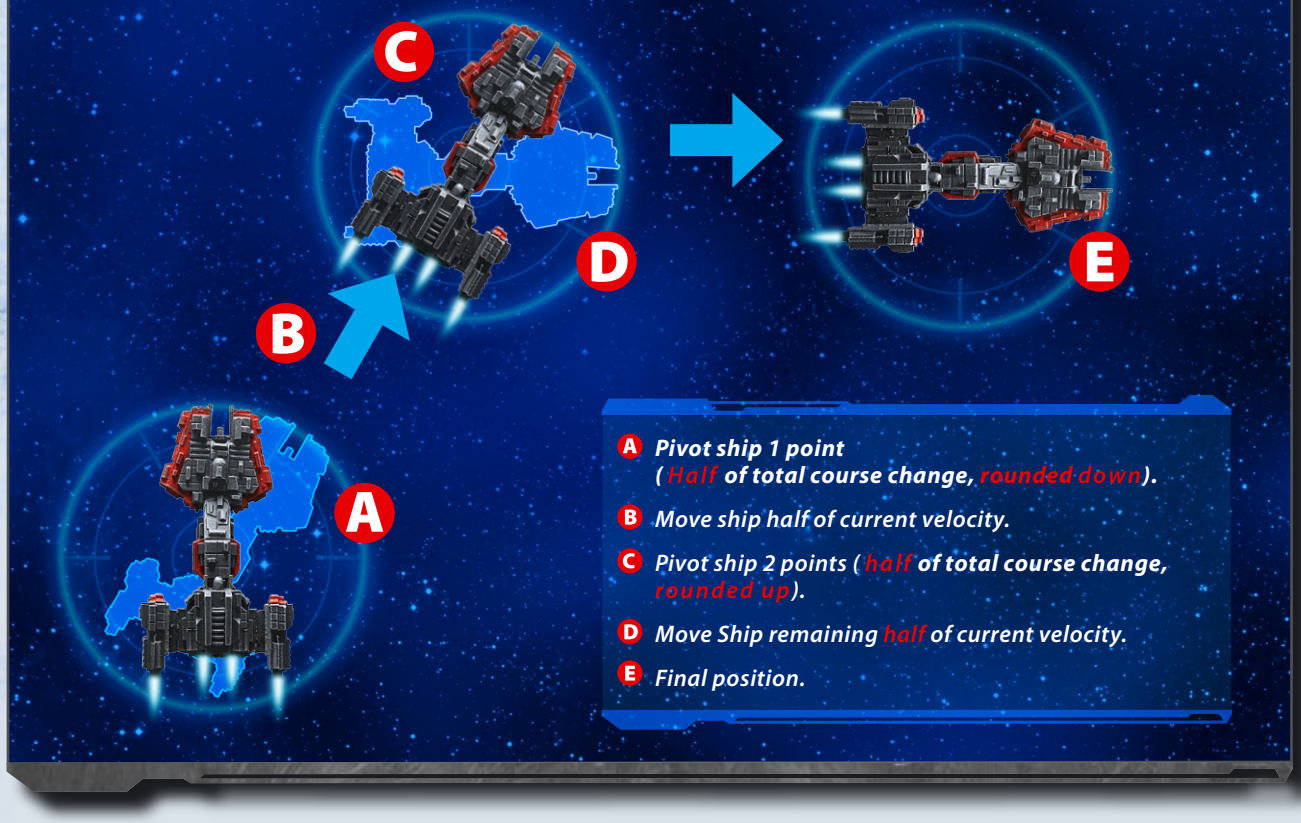
ACCELERATION AND DECELERATION:

Once a ship is moving, it will continue to move at the same velocity every turn unless **THRUST** is applied to change that velocity. Thrust points may be used to speed the ship up (acceleration) or slow it down (deceleration), with each thrust point changing the ship's velocity by 1mu up or down. The use of thrust for acceleration or deceleration is written in the order plotting as "plus" for acceleration thrust or "minus" for deceleration, and the number of thrust points being used – for example, an order written as "+2" would be using 2 thrust points for acceleration (adding 2mu to the speed), while "-3" would be three points (3mu) of deceleration. Any velocity change is applied in the same turn that it is plotted, so if a ship ended the previous turn at velocity 6, and has a +2 acceleration plotted for the new turn, it will move 8mu this time. Ships must **ALWAYS** be moved the full and exact distance indicated by their current velocity – the only way to slow a ship down is to apply decelerating thrust. Ships may **NOT** move backwards.

Each thrust point changes ship velocity by 1 mu. Ships must always move full distance indicated by current velocity. Ships may not move backwards.



EXAMPLE OF A 3 POINT COURSE CHANGE TO STARBOARD (S3)



CHANGING COURSE:

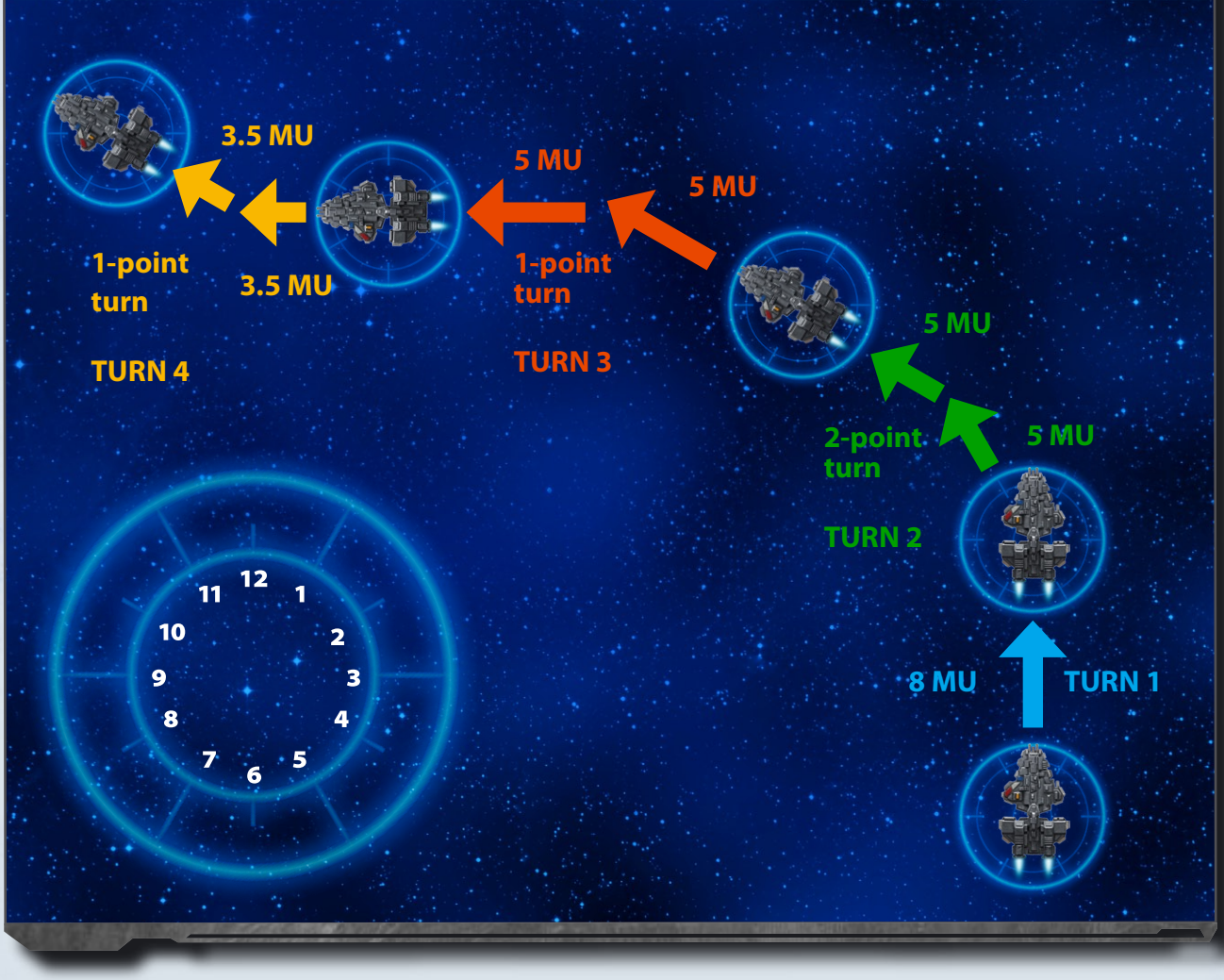
Thrust points may also be used to change the course of the ship, at a rate of 1 thrust point = 1 course point (30 degrees) of turn. Course change orders are written in the order plotting in a similar way to velocity changes, with either a "P" (for turns to Port, or anticlockwise) or an "S" (for Starboard, or clockwise) followed by the number of thrust points applied to the course change. In terms of the "clockface" course references, Port turns are "minus" and Starboard turns are "plus" – eg: if a ship is currently heading on course 3, then a 2-point turn to Port (written P2) would turn the ship onto course 1, while a 2-point turn to Starboard (S2) would turn it to course 5. Turning the miniature through the correct angle is a lot easier if a simple circular "turn gauge" is cut from card and marked with the 12 clockface points , but it can also be judged accurately enough from the hexagonal base of the ship's stand. Whenever a ship changes course, the change is performed half at the **START** of the ship's movement for that game turn, and half at the **MID-POINT** of the movement.

⚠ **EXAMPLE:** ***To explain this more fully, actual movement of the ship miniature is performed like this: first the miniature is turned (pivoted on the spot) by HALF - rounded DOWN - of the total amount of course change plotted for that game turn. Now the miniature is moved HALF its total velocity straight forward in the direction it currently points. Then it is pivoted the remaining half - rounded UP - of its total course change, and finally moved the remaining half of its total velocity. Note that because of the rounding down of the first part and up of the second part of the any course change, if the ship is only changing course by 1 point then that will occur at the MID-POINT of the movement, with none at the beginning.***

⛔ *Each thrust point turns ship by 30 degrees (1 clockface point). Half (round down) of turn at starting point, remaining half after moving half its velocity.*

ORDER PLOTTING EXAMPLE

SHIP ID	START V	START C	TURN 1	V	C	TURN 2	V	C	TURN 3	V	C	TURN 4	V	C
Example	6	12	+2	8	12	+2, P2	10	10	P1	10	9	-3, S1	7	10



FIRE PHASE:

In this phase all ships that can acquire valid targets may fire any or all of their weapons.

In the basic game any given ship may fire any or all of its weapons systems, provided there is a target in range and in the weapon's firing arc; as a general rule, each weapon may fire once only in each game turn. The number of different targets that any one ship may engage is determined by its available **FIRE CONTROL SYSTEMS (FCS)**, as explained below. Within the limitations of the **FCS** rules, a ship's weapons fire may be divided between available targets in any way the player desires. The fire phase starts with a die roll for **INITIATIVE** between the players - each rolls, and the highest scorer has the Firing Initiative for that phase - that player may select any **ONE** of his ships and perform **ALL** desired firing from that ship against any available targets. Once the player declares that all firing from that ship is completed, then the opposing player may select one of his own ships and perform all desired firing from it. Firing then alternates between the players on a ship-by-ship basis until all desired firing is completed. Note that because all damage is inflicted immediately it is possible for a ship to be destroyed (or to lose important weapons systems) **BEFORE** it has the chance to fire back in that game turn.

Roll for initiative, winner selects first ship to fire. Resolve fire from that ship and apply damage effects to target immediately. Players alternate firing ships until all firing is completed.

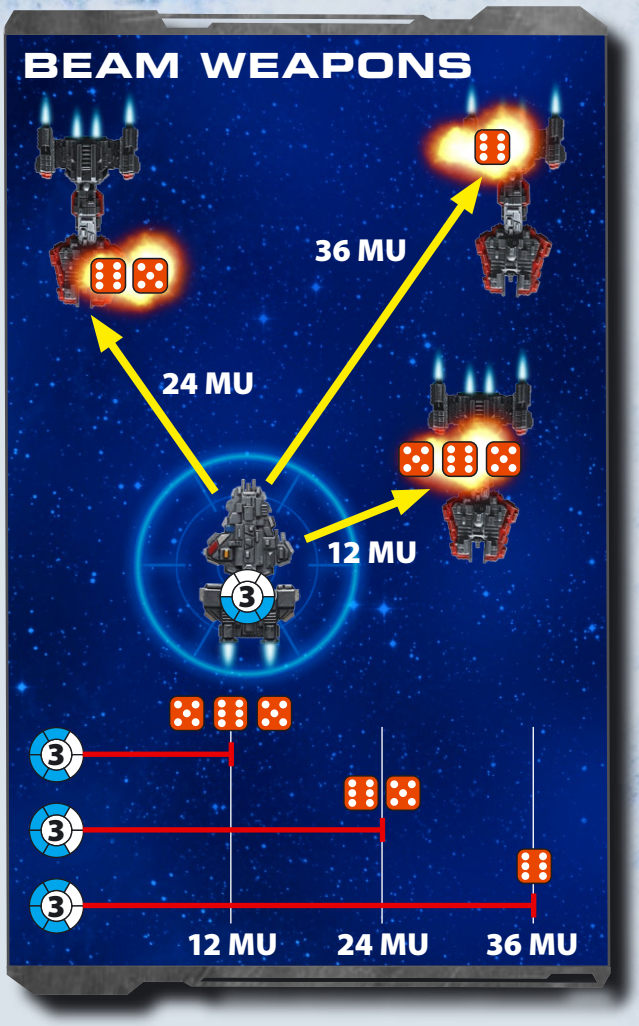
FIRE CONTROL SYSTEMS:

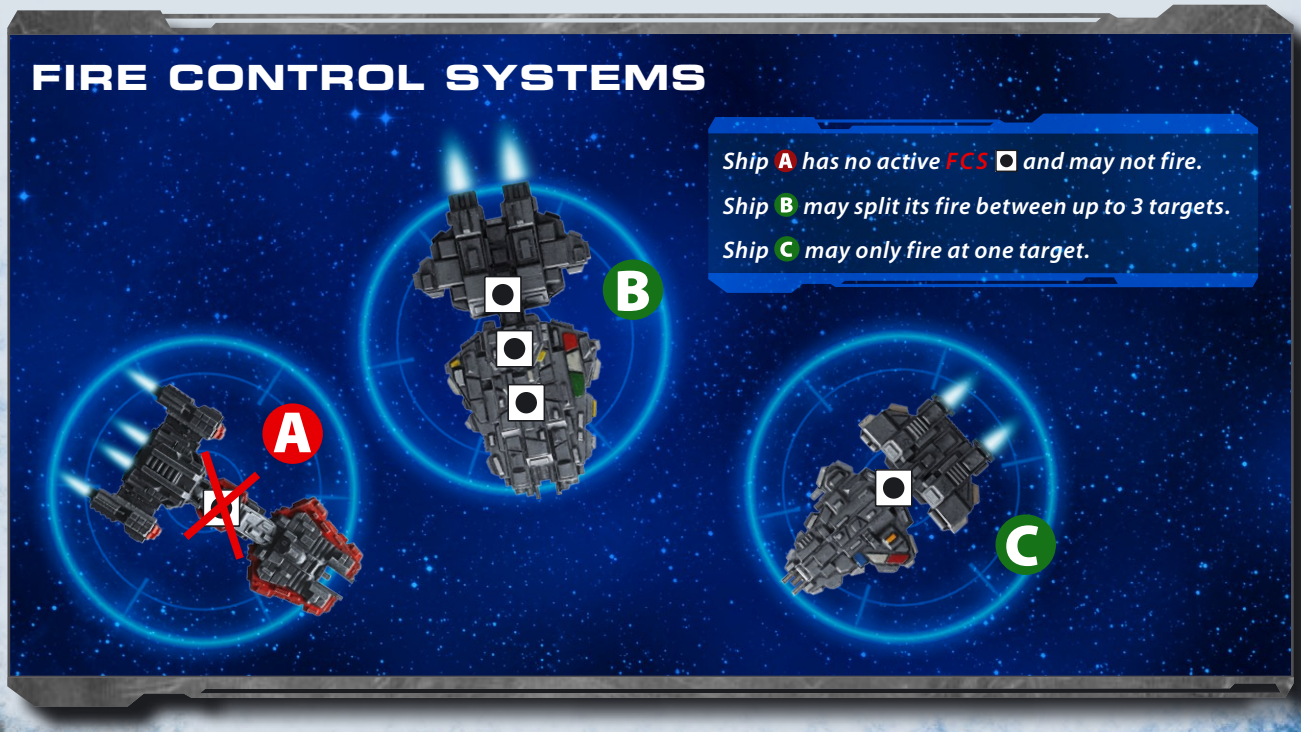
Each ship must have at least one functioning Fire Control System (FCS) in order to fire its weapons. Each FCS icon represents the ability to engage **ONE** target ship in any game turn - for example a Frigate, with only one FCS, can only fire at one target per turn - but it may fire any or all of its weapons at that target, provided that they are within range and firing arc. A Battlecruiser with three FCS can split its fire between **UP TO THREE** different target ships in the same turn, dividing its available weapons in any combination between the targets provided that they are within range and fire arc. If a ship loses all its FCS due to damage, it may not fire even if it still has operational weapons.

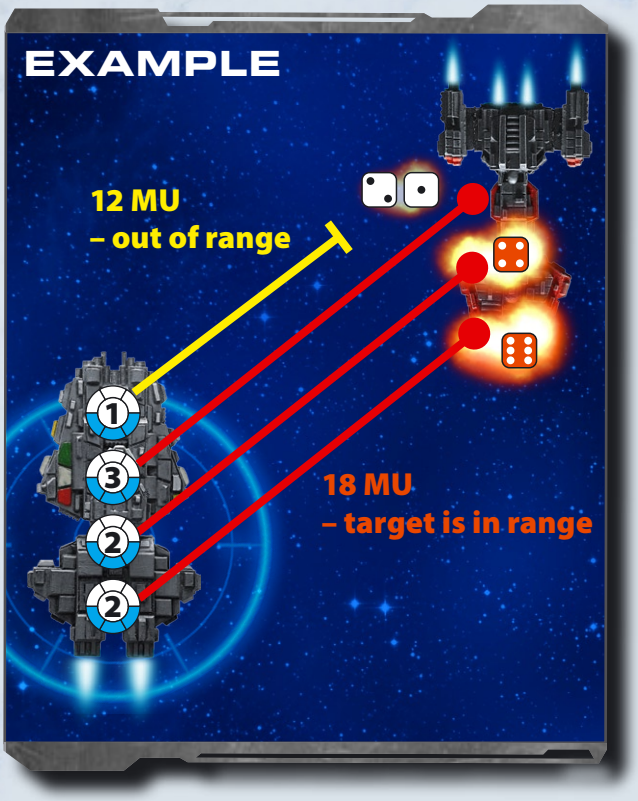
Ships may engage one target for each operational Fire Control System

FIRING BEAM WEAPONS:

Each Beam Weapon icon on a ship represents an energy beam projector, which is the primary weapon system used in Full Thrust Light. The number inside the icon is the "class" of the beam system, from the smallest (CLASS 1) upwards. Each beam weapon on the ship may fire once per game turn, at one target ship, provided that it is within range







and fire arc. At up to 12mu range, a beam weapon rolls a number of dice equal to its **CLASS**; in every further 12mu range band, this drops by one die. Thus, a CLASS 1 Beam rolls just ONE die at up to 12 mu (its maximum range); a CLASS 2 rolls two dice at up to 12 mu and one die out to 24 mu; a CLASS 3 rolls three dice up to 12 mu, two dice up to 24 mu and just one up to 36 mu. For each die rolled, damage is inflicted on the target as follows: Scores of 1, 2 or 3 = no effect (a miss); 4 or 5 = 1 damage point inflicted, 6 = 2 damage points inflicted. All the beam weapon fire from one ship against a single target ship may be rolled together.

⚠ **EXAMPLE:** ***A Cruiser is firing at a target 18 mu away using two Class 2 beams (1 die each at 12-24mu) plus a single Class 3 beam (2 dice at 12-24mu), then the player may roll the total of 4 dice at the same time. If he rolled 1, 2, 4 and 6, he would score a total of 3 points of damage (1 point with the roll of 4, 2 points with the 6; the 1 and 2 miss). The cruiser cannot fire its class 1 beams because the target is out of range for them.***

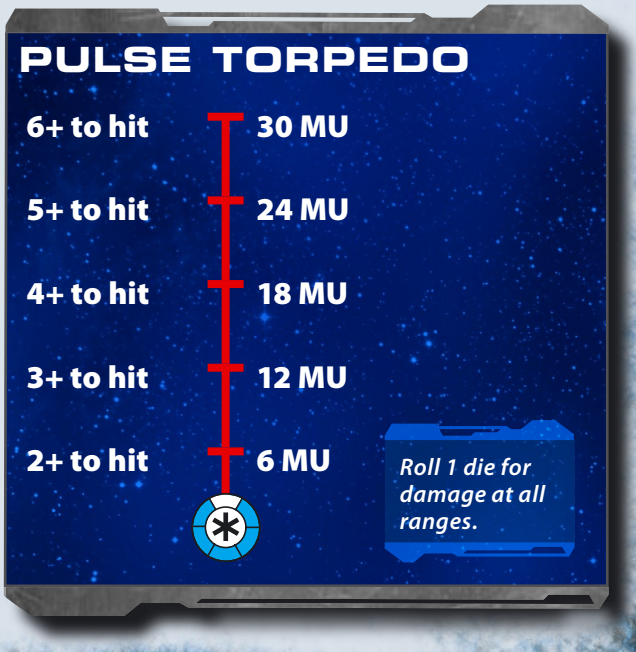
⛔ *Beam Weapons roll number of dice according to class at 12 mu, reduce by one die per additional 12 mu. Damage is 1 if score is 4 or 5; 2 if score is 6.*

FIRING PULSE TORPEDOES:

Pulse Torpedo Launchers are "heavy weapons" with more punch than beam systems; they are generally fitted to the larger ship classes, though specialized smaller ships with PT mountings (and little else) are also possible. The icon for a PTL is similar to the beam icon but with an asterisk (*) inside the circle in place of the beam class number. Each Pulse Torpedo Launcher on the ship may fire once per game turn, at one target ship, provided that it is within range and fire arc. Firing a PTL is a two-step process - first a **ROLL TO HIT**, according to the range to the target: a single die is rolled per PT fired: at up to 6mu a score of 2+ is needed to hit, and this rises by 1 for every extra 6mu of range - so at 6-12mu it is 3+, 12-18mu 4+, 18-24mu 5+ and finally at 24-30mu a 6 is needed. The maximum range of the Pulse Torpedo is 30mu. If the PT hits the target, then a second die is rolled - the actual number rolled (1 to 6) is the number of damage points inflicted on the target ship.

⚠ **EXAMPLE:** ***A Battlecruiser fires two PT launchers at a target ship 20mu away. This is in the 18-24mu band, so 5+ is needed to hit. The player rolls a 4 and a 6, so one PT misses and one hits. Now the player rolls the damage die for the hit, and scores 3 - the PT hit inflicts 3 points of damage to the target.***

⛔ *Pulse Torpedoes need to score 2 to hit at 6mu, add 1 to needed score per additional 6 mu. If a hit is scored, roll again, actual number rolled is damage.*



RECORDING DAMAGE:

Damage points inflicted are applied to the target ship immediately, and are recorded by crossing off the small square **HULL BOXES** on the ship's SSD, starting at the top row and working from left to right, one box per damage point. When the accumulated damage reaches or passes the end of one of the hull box rows, then a **THRESHOLD CHECK** for critical systems damage is performed as soon as all fire by that particular attacking ship against that target ship is completed. To perform a threshold check, one die is rolled for **EACH** system icon on the ship. At the **FIRST** threshold (first row of hull boxes all marked off), systems are knocked-out on rolls of 1 only; one the **SECOND** threshold (second row marked off) systems are lost on scores of 1 or 2, and at the **THIRD** threshold (third row) 1, 2 or 3.

If **MORE** than one threshold point is passed in the attack from a single ship (which can occur, especially with smaller target ships), then only roll for the "worst case" threshold but add 1 to the threshold number - eg: if a ship took so much damage in one attack that both the first and second rows of hull boxes were lost at the same time, only the second row threshold would be rolled for but it would be treated as if it were a third-row check, with systems lost on 1, 2 or 3 instead of just 1 or 2. Systems knocked-out on a threshold check are crossed off the SSD and may not function again for the rest of the game.

When a **DRIVE SYSTEM** icon is lost, the ship loses the use of the thrust points provided by that drive (which may be all or part of the ship's total available thrust, according to whether the ship has more than one drive icon or not). When **ALL** of a ship's Hull Boxes are marked off, then the ship is destroyed and removed from play.

⚠ **EXAMPLE:** ***An undamaged Heavy Cruiser is hit by enemy fire, taking 6 damage points. The first row of hull boxes is marked off, requiring a threshold roll for each system on the SSD. A die is rolled for each system, with a roll of 1 resulting in destruction of that system.In our example, a Fire Control System and a Beam Weapon are lost.***

Later in the game, the damaged cruiser takes another hit from the enemy, this time 8 damage points. Two more hull rows are now marked off and another threshold roll is required. At the third threshold, systems would normally be lost at a roll of 1, 2 or 3. As the damage passed two thresholds in our example, 1 is added to the roll. This means that systems will be destroyed at rolls of 1, 2, 3 or 4. Our Cruiser is badly shot up and can no longer change course or velocity but can still fire its remaining Beam Weapons because it still has a functioning Fire Control System.

⛔ *Cross off hull boxes from left to right, top to bottom row. Threshold roll is required if end of row is reached. Systems are lost on roll of 1 at first threshold, 2 or less at second, 3 or less at third. If multiple rows are marked off, check only for the highest threshold but add 1 to the required roll*





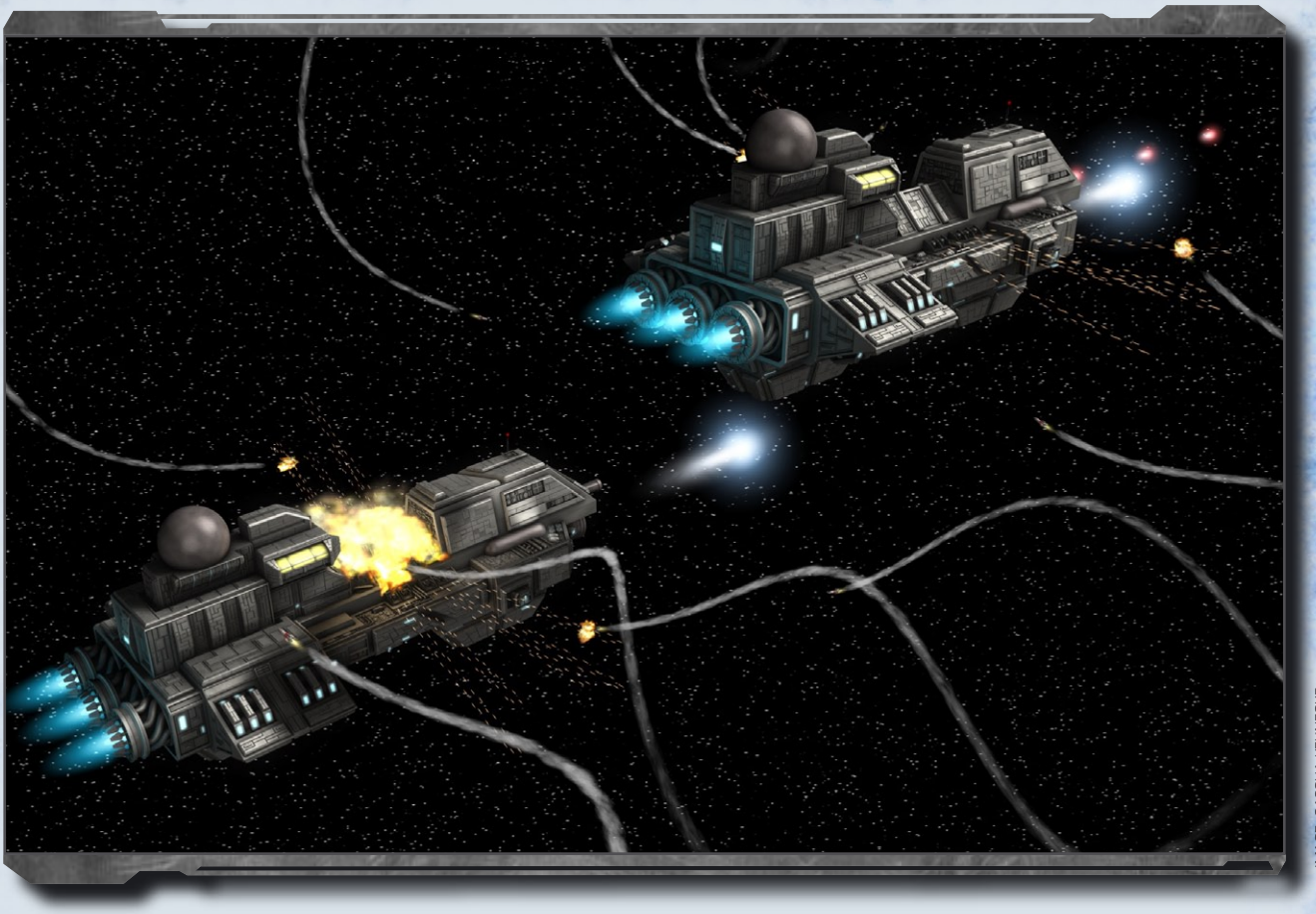


IMAGE: SASCHA KEUNECKE

ENDING THE GAME:

The game ends when one player's fleet is defeated. Whether this means complete destruction of all ships, or whether one player decides he has lost and withdraws his surviving ships from the battle area, is up to the players to define. Specific scenarios are beyond the scope of this introductory rule set, but it is always a lot more interesting to set some victory conditions rather than just fight to the last ship - a simple objective like protecting or attacking a freighter or convoy will make a very enjoyable game.

Game ends when one side is destroyed, withdraws or victory conditions are met.

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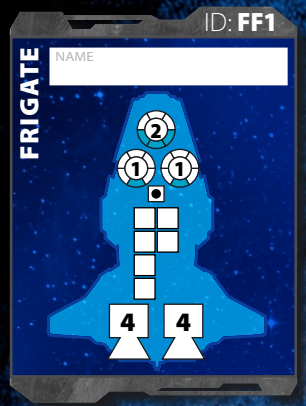
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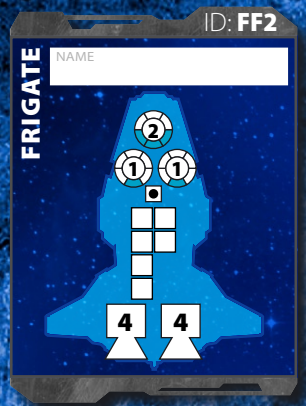
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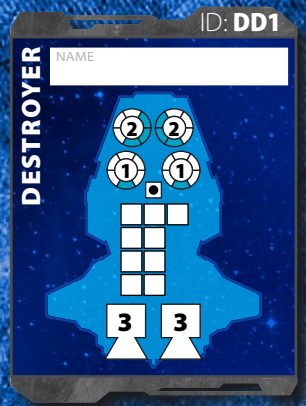
EDITORIAL STAFF: Christian Weinhold, Sascha Keunecke

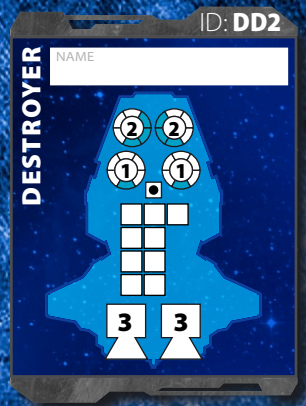
GRAPHICS, TYPESETTING AND LAYOUT:
Stephan Ludewig

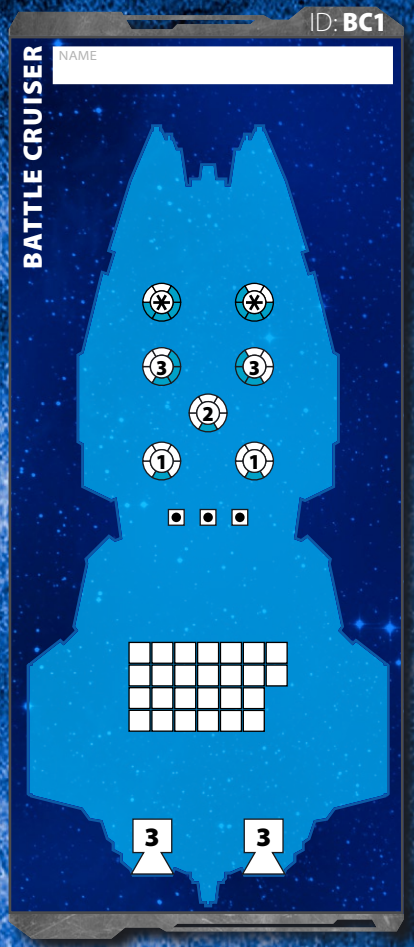
CG ARTWORK: Sascha Keunecke, Stephan Ludewig

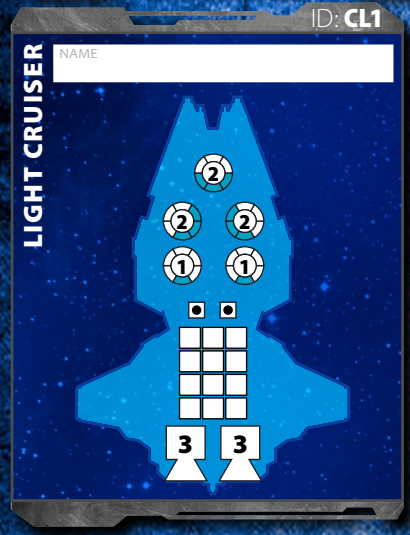


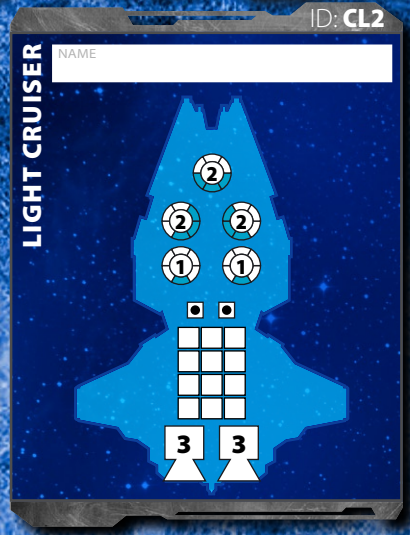


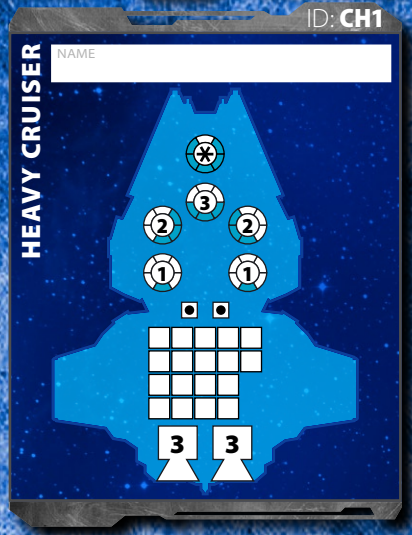


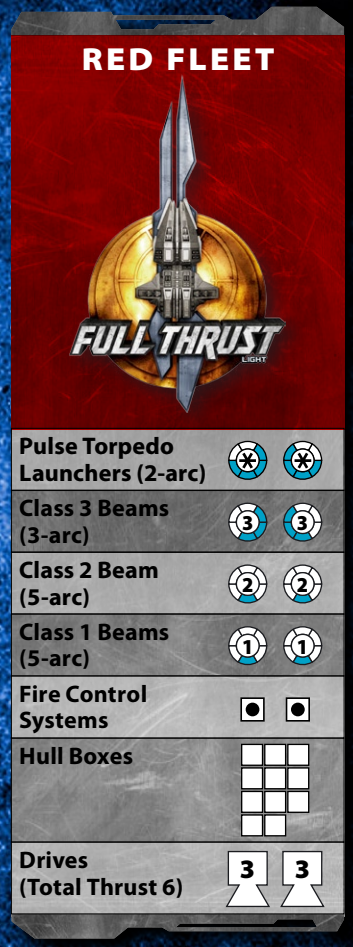




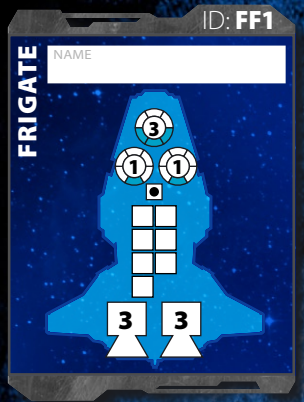


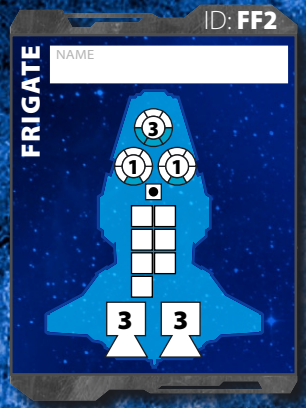


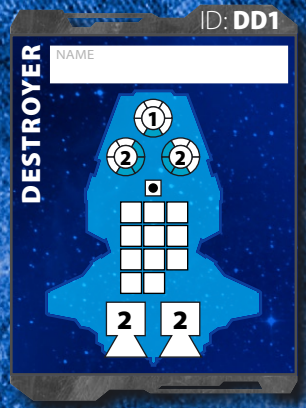


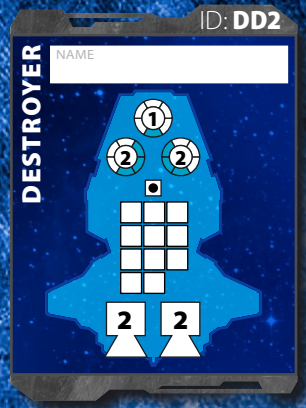


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Example	6	12	+2	8	12	+2, P2	10	10	P1	10	9	-3, S1	7	10	–	7	10	S3	7	1	+5, S1	12	2	+6	18	2	-3, P3	15	11
FF1																													
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DD1																													
DD2																													
CL1																													
CL2																													
CH1																													
BC1																													



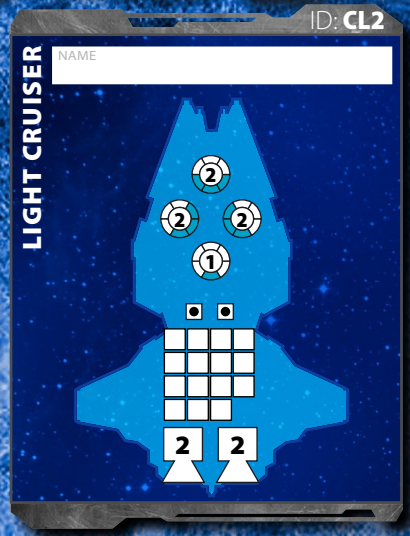


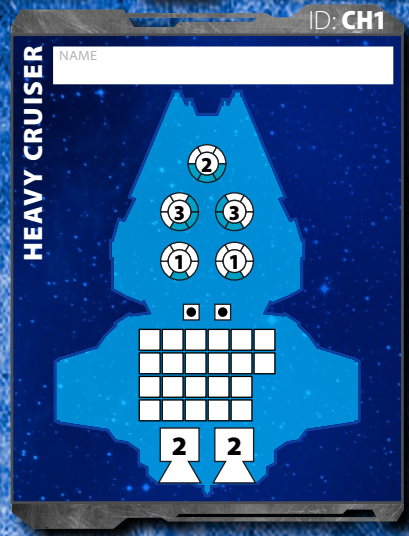


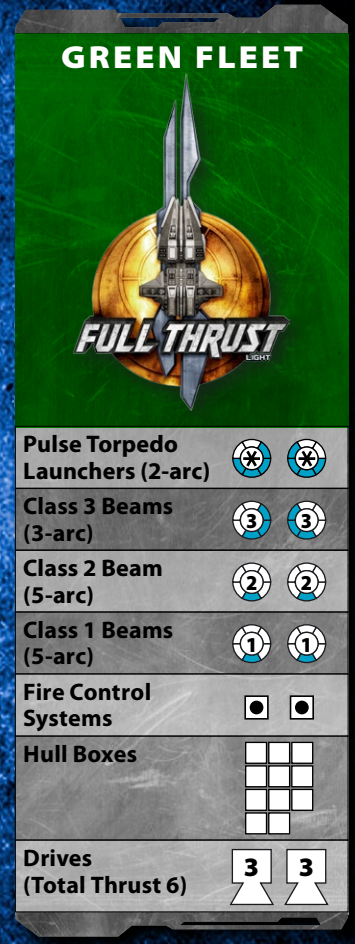












SHIP ID	START V	START C	TURN 1	V	C	TURN 2	V	C	TURN 3	V	C	TURN 4	V	C	TURN 5	V	C	TURN 6	V	C	TURN 7	V	C	TURN 8	V	C	TURN 8	V	C
Example	6	12	+2	8	12	+2, P2	10	10	P1	10	9	-3, S1	7	10	–	7	10	S3	7	1	+5, S1	12	2	+6	18	2	-3, P3	15	11
FF1																													
FF2																													
DD1																													
DD2																													
CL1																													
CL2																													
CH1																													
BC1																													